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SEQUENCE AND SERIES

Exercise-1 : Single Choice Problems

1. If a, b, c are positive numbers and $a+b+c=1$, then the maximum value of $(1-a)(1-b)(1-c)$ is :

(a) 1 (b) $\frac{2}{3}$ (c) $\frac{8}{27}$ (d) $\frac{4}{9}$

2. If $xyz=(1-x)(1-y)(1-z)$ where $0 \le x, y, z \le 1$, then the minimum value of $x(1-z)+y(1-x)+z(1-y)$ is :

(a) $\frac{3}{2}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{1}{2}$

3. If $\sec(\alpha-2\beta)$, $\sec\alpha$, $\sec(\alpha+2\beta)$ are in arithmetical progression then $\cos^2\alpha=\lambda\cos^2\beta$ $(\beta \ne n\pi;\ n \in I)$ the value of λ is :

(a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$

4. Let a, b, c, d, e are non-zero and distinct positive real numbers. If a, b, c are in A.P. ; b, c, d are in G.P. and c, d, e are in H.P., then a, c, e are in :

(a) A.P. (b) G.P.

(c) H.P. (d) Nothing can be said

5. If $(m+1)^{\text{th}}$, $(n+1)^{\text{th}}$, and $(r+1)^{\text{th}}$ terms of a non-constant A.P. are in G.P. and m, n, r are in H.P., then the ratio of first term of the A.P to its common difference is :

(a) $-\frac{n}{2}$ (b) $-n$ (c) $-2n$ (d) $+n$

6. If the equation $x^4-4x^3+ax^2+bx+1=0$ has four positive roots, then the value of $(a+b)$ is :

(a) -4 (b) 2

(c) 6 (d) can not be determined